Covered Calls Around M&A Agreement Disclosures

Solo submission GQH Systematic Trading / Massive 8-K Options

Hypothesis and mechanism. Writing an out-of-the-money call may add more net value to synthetic stock after M&A agreement filings than on ordinary dates for the same names:

$$\theta = \mathbb{E}[(CC - S)_{\text{net}} \mid \text{event}] - \mathbb{E}[(CC - S)_{\text{net}} \mid \text{ordinary}] > 0.$$

Strategic-upside demand could support premiums [1]. Genuine deal risk and repricing before entry compete with this explanation. Covered calls contain equity, volatility and timing exposures [2]. The literature motivates the test; it does not establish an M&A premium or stock rally.

Method and provenance. V4 uses static TOP100 and Massive's `acquisition_agreement` or `merger_agreement` tags, deduplicated by accession; ambiguous filers are skipped. Tags do not prove buyer status, control, first announcement or unique transactions. The preceding-filing-session chain selects 90–180 DTE, nearest 120, approximately 5% OTM. Entry is the next NYSE session close; exit is entry plus 21 sessions, capped at expiry. Synthetic stock is ATM call minus put plus strike discounted at flat 4% carry.

The screened-buyer hypothesis was committed as `63664eb` before first IS. V4 changes selection/timing *after earlier IS inspection*; it is not the original preregistration. IS is 2024–2025; January–August 2026 OOS was evaluated once with unchanged source. Judges alone evaluate sealed dates. Historical hypotheses and V1–V4 runs remain disclosed.

Each option leg costs 5% of premium per side; double costs are a fixed stress. Up to 120 seed-7 ordinary draws lie over 30 calendar days from agreement tags. No price-based redrawing. Missing or over-three-session-stale marks remain unavailable.

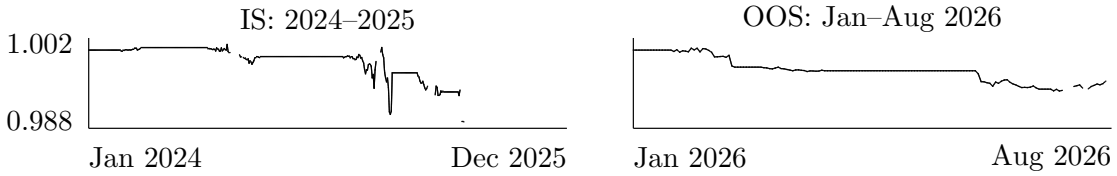
Fixed primary result. IS $\theta = 0.87$ pp, exploratory 95% CI $[-0.37, 2.17]$; OOS 3.38 pp, CI $[1.18, 5.85]$. IS has 56 accessions/25 issuers and 38 finite increments; OOS 25/14 and 15 finite increments. OOS matched stock/covered-call means are $-4.50/-3.38\%$ at events versus $1.79/-0.48\%$ on ordinary dates. Positive θ is *not profitable covered-call performance*: short calls cushion falling event stocks while limiting rising controls. It does not establish mispricing or causality. Table units are percentage points; counts are finite event/ordinary observations.

	IS		OOS	
Horizon	N_E/N_O	θ [95% CI]	N_E/N_O	θ [95% CI]
1	45/92	0.22 [-0.10, 0.57]	21/103	0.18 [-0.63, 0.86]
2	44/89	0.26 [-0.18, 0.68]	21/99	0.66 [-0.21, 1.54]
3	45/90	0.48 [-0.09, 1.13]	20/98	1.02 [0.14, 1.94]
5	44/89	0.78 [-0.02, 1.62]	19/94	0.73 [-0.48, 2.08]
10	40/90	1.35 [0.39, 2.38]	19/93	1.61 [0.32, 3.08]
21	38/84	0.87 [-0.37, 2.17]	15/86	3.38 [1.18, 5.85]
42	31/71	-1.93 [-5.92, 1.03]	15/80	5.56 [2.21, 9.16]
63	34/61	0.95 [-2.75, 4.52]	13/57	4.04 [-1.04, 9.34]
Expiry	33/59	0.49 [-4.30, 5.01]	11/50	2.12 [-6.07, 8.60]

Robustness and interpretation. Doubled-cost θ: IS 0.95 pp (CI $[-0.39, 2.34]$), OOS 3.51 pp ($[1.21, 6.07]$). All 27 close-entry/21-session expiry–OTM–tag point estimates are positive in each window; all 972 correlated cells per window are retained, without choosing a winner. Pre/filing-day views are diagnostic. OOS all-issuer leave-one-out estimates range 2.08–4.15 pp. Row bootstrap ignores overlapping deals/issuers: OOS effective issuer counts are 9.8 events/7.0 controls. Evidence supports the relative increment in this proxy sample, not a stable profitable edge. No OOS retuning.

Capital, risk and trading. Each window starts at $1 million with no carried positions: one contract per signal, one concurrent per name, zero cash return, no leverage. Stock-reference name/industry/total limits are 10/25/100%; pause/resume drawdown 10/5%, using prior-session state. The inherited 14-name industry map makes this a different sample from the event study. IS has one unpriced entry and one unresolved exit; OOS has two open positions at the window end.

Calendar portfolio, net	IS	OOS
Total return / CAGR / volatility / Sharpe	NA	NA
Full-path maximum drawdown / turnover	NA	NA
Observed drawdown (partial path)	1.20%	0.66%
Known stock-notional / initial capital	0.7046×	0.1295×
NAV gaps / sessions	130/502	6/166
Commitments / fully priced closes	13/11	5/3



Curves show NAV/initial capital; gaps are unfilled. Missing terminal marks prevent complete-path metrics and factor regression. The IS notional is an observed subtotal; neither initial-capital ratio substitutes for formal turnover. All-three-leg positive same-day volume covers 36/51 IS and 17/21 OOS priced events. One lot/minimum-leg volume is median 9.55/3.45%, maximum 100/33.33%: this is not depth or fill evidence. No scalable dollar capacity is established [4]. Stock downside and capped upside remain. Standard 100-share deliverables are assumed; commissions, impact, dividends, assignment and financing spreads are not reconstructed.

Limits and reproduction. Static-universe survival, unproven historical tag availability, repeated transactions and nonrandom coverage limit inference. News may precede filing [3]. OOS pre-entry implied/absolute-realized medians are 8.45/7.28% (15 events) and 8.54/5.24% (87 controls); this yardstick alone proves no premium. The notebook takes dates and a Massive key; inputs stay private. AI assisted coding/drafting; the author retains responsibility. Sealed performance is unknown.

Sources. Data/code: Massive, GQH 8-K Options starter and track rules. [1] Gârleanu et al. (2009), *Demand-Based Option Pricing*, RFS 22. [2] Israelov and Nielsen (2015), *Covered Calls Uncovered*, FAJ 71. [3] Ben-Rephael et al. (2022), *Who Pays Attention to SEC Form 8-K?*, TAR 97. [4] Christoffersen et al. (2018), *Illiquidity Premia in the Equity Options Market*, RFS 31.